MASTER'S MODULE · AGENTIC AI · WINTER SEMESTER

Evaluation, Accountability & Footprint

Does the agent actually work — and at what cost, to whom? Closing the loop on a semester of agentic AI.

Unit 09 · Part 2 of 2 · Session 14 · Week 14
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Today's agenda

1. **Why evaluation is hard** nondeterminism, many valid trajectories, compounding errors
2. **LLM-as-judge** rubrics, pairwise comparison, the judge's biases and their fixes
3. **Benchmarks & your own evals** SWE-bench, GAIA, tau-bench; eval sets from traces; CI; the Lab 14 harness
4. **Accountability** who answers when an agent acts; EU AI Act, GDPR, audit trails
5. **Footprint & wrap-up** energy and cost; the nine units in one picture; the exam

Where we are

Unit 09, part 2 — the final session of the course

Last week (S13) — we learned to attack agents and to defend them — injection, exfiltration, containment

The whole semester — thirteen labs on a local Ollama model: loop, tools, memory, RAG, sandbox, traces, security

The last question — does it actually work — reliably, accountably, affordably?

Today — evaluation methods, legal accountability, energy and cost footprint

And then — the nine units in one picture, and what the exam expects of you

GUIDING QUESTION

Security asked: what can go wrong? Evaluation asks the harder everyday question: is it even going right?

1

Why agent evaluation is hard

Nondeterminism, trajectory multiplicity, compounding errors, expensive rollouts

Why your test suite does not transfer

Nondeterminism — the same task samples different runs; one pass proves existence, not reliability

Many valid trajectories — ten correct ways to research a topic — exact-match on steps is meaningless

Compounding errors — illustration: 20 required steps, each 0.95 success given all earlier successes, no recovery: 0.95^20 ≈ 35.8%

- a deliberate simplification: real agents have dependent steps and recovery paths

Expensive rollouts — one eval case is a full agent run: minutes and paid tool calls, not microseconds

Moving substrate — model upgrades silently shift behaviour under an unchanged prompt and eval set

CONSEQUENCE

Agent evaluation is statistics over repeated runs, not assertions about one run. Sample, aggregate, track over time.

The evaluation pyramid

Four levels, rising in cost and falling in determinism



- Run the cheap left levels constantly; reserve full rollouts for release gates.
- Outcome checks inspect the actual environment or artifact, never just a final message that claims success.
- Trajectory checks start with deterministic invariants: action policy, allowlists, step and token budgets.

Trajectory vs final-outcome evaluation

Two questions about the same run — diagnosis vs verdict

Trajectory: judge the path

- were steps sensible given the state?
- efficiency: detours, repeated calls, loops
- safety: any policy-violating action en route?
- deterministic invariants first: policy, budgets
- reference traces or a judge only for the rest

Outcome: judge the result

- did the artifact or environment meet the goal?
- inspect the world, not the final message
- often checkable: tests pass, file or state is right
- cheap to grade, but a lucky fragile path passes
- use both: outcome gates, trajectory diagnoses

2

LLM-as-judge

Scaling judgment — and inheriting the judge's biases

LLM-as-judge: the mechanics

The idea — a strong LLM grades outputs no exact check can score (Zheng et al., 2023)

Rubric scoring — absolute 1-5 per criterion: completion, groundedness, efficiency, safety

Pairwise comparison — A versus B on the same task — often steadier than absolute scores, not always

Reference-guided — hand the judge a gold answer or trace; grading sharpens markedly

Why it scales — judging is usually cheaper than producing — but it is still a fallible model call

CALIBRATE FIRST

Calibrate on representative, stratified cases: human agreement plus error analysis. Grow the sample with risk and the precision you need. No calibration, no judge.

The judge's biases — and their mitigations

A judge is a model: it carries systematic, measurable distortions

Bias	What you observe	Mitigation
Position bias	first-listed answer wins regardless of content	swap positions; count inconsistent verdicts separately, not as ties
Verbosity bias	longer, list-heavy answers score higher	control length; rubric rewards economy
Self-preference	judge favours its own family's style	different model family reduces it; verify on labels
Leniency drift	scores cluster high; everything gets 4/5	pairwise instead of absolute; anchor examples
Shallow verification	fluent but wrong reasoning passes	supply reference answers; decompose into checks

Catalogue from Zheng et al. (2023) and follow-ups. Every mitigation costs extra calls — judging is engineering, not magic.

A pairwise judge that audits itself

Five outcomes, not one tie: report win rate with coverage, invalid and inconsistent rates; re-judge hard cases

judge.py

```
VERDICT = {"type": "string",
           "enum": ["first", "second", "tie"]}
def ask(task, a, b):
    p = JUDGE.format(task=task, a=a, b=b)
    try:  # structured output: the enum is enforced
        return llm(p, response_schema=VERDICT)
    except (ValueError, TimeoutError):
        return "invalid"  # error path stays visible

def judge_pair(task, a, b):
    v1, v2 = ask(task, a, b), ask(task, b, a)  # swapped
    if "invalid" in (v1, v2): return "invalid"
    if (v1, v2) == ("first", "second"): return "A"
    if (v1, v2) == ("second", "first"): return "B"
    if (v1, v2) == ("tie", "tie"): return "tie"
    return "inconsistent"  # first/first: position bias
```

3

Benchmarks & your own evals

Public leaderboards, honest readings, and the eval set you own

The agent benchmark landscape

What each one actually measures — and its honest caveat

Benchmark	What it measures	Score unit	Honest caveat
SWE-bench Verified (Princeton / OpenAI, 2024)	patch real GitHub issues until tests pass; 500 human-validated tasks	% issues resolved	score depends on harness and inference budget; cite a dated leaderboard snapshot
SWE-bench Pro (Scale AI, 2025)	harder, longer-horizon issues; held-out commercial split	% issues resolved	a separate benchmark, not the same measurement series as Verified
GAIA	general assistant tasks: web, files, tools	exact-match accuracy	answers circulate; harness quality dominates scores
AgentBench	eight interactive environments: OS, DB, web, games	per-env success	heterogeneous, ageing fast, rarely reported in full
tau-bench (tau2)	tool use plus user dialogue under policy	pass^k reliability	simulated users; retail, airline, since tau2 (2025) telecom with dual control

IID sketch, per-trial success p: pass@k = 1-(1-p)^k (any of k), pass^k = p^k (all k). p=0.8, k=5: 99.97% vs 32.77% (Yao et al., 2024).

Reading leaderboards honestly

Contamination — public tasks and answers may sit in training corpora — a risk to check per model, not a given

Verified subsets — SWE-bench Verified exists because many originals were underspecified or unsolvable

Versions and budgets — scores depend on benchmark version, harness and inference budget; never compare headline % across variants

Scaffold vs model — entries compare harness plus model; scaffolding alone is worth tens of points

Benchmarketing — vendors quote the one benchmark, scaffold, and k where they win

Transfer gap — a high SWE-bench score says little about your support agent

RULE

A benchmark score is a claim about a benchmark. Only your own eval set makes claims about your system.

Your eval set: from traces to CI gate

The S11 tie-back: production traces are eval cases waiting to be curated

Source — production traces (S11) plus the small baseline task set you have kept since S02

Curate — deduplicate, anonymise, attach outcome or rubric; split dev, regression and untouched holdout

Grade — exact checks first, judge rubrics only where needed; a safety violation fails the case outright

Size — start small and stratified, grow with risk; report n, repeats and confidence intervals

CI gate — every prompt, tool or model change; freeze model, tools, environment and budget per comparison

Version — eval sets are code: review, version, track score and cost per successful task over time

PROMPTS ARE CODE

A prompt change ships with passing evals, or it does not ship. Same for a model swap.

Lab 14: the evaluation harness

Offline and synthetic: benchmark tasks, corpus and thirty pre-recorded agent runs in data/; judge on local Ollama

Part B: outcome checks

Mechanical gates before any judge: report format, every [D..] citation resolves to a real corpus document, groundedness. R4: complete check_citations.

Part C: trajectory checks

Deterministic invariants over the recorded traces: fetch before drafting, step budget, no repeated searches, URL allowlist. R1: why outcome-only fails.

Part D: self-auditing judge

Pairwise LLM-as-judge with the swap test; defences against position, verbosity and self-preference bias; calibrated on ten hand-labelled reports and five pairwise labels. R2: adopt it?

Part E: footprint accounting

Tokens, toy cost and a rough energy figure per run from the span token counts; the multiplier between padded baseline and lean variant. R3.

Part F: regression gate

Mean judge score over three trials per variant and task; regression_ok fails when groundedness drops beyond tolerance. R5.

Part G and wrap-up

Tune tolerance and step budget; then fourteen labs in one table. Gap-fill cells with fold-out solutions; report tasks R1 to R5; 90 to 120 min.

4

Accountability & Footprint

Who answers when an agent acts, what it costs — and the semester in one picture

Who answers when an agent acts?

Not the agent — software has no legal personhood; responsibility lands on humans and organisations

Provider — two roles: the model provider (GPAI) and the system provider; documentation, risk management, disclosed limits

Deployer / operator — runs it for a purpose: correct use, oversight, monitoring, logging

User — can be contributory — ignoring confirmations, off-label instructions

Today's reality — duties follow the applicable law, actor role and use case; contracts add duties, never replace them

DESIGN LINK

Approval gates (S10) place a named human in the causal chain and document who decided. They do not shift responsibility by themselves.

EU AI Act (as of 2026), GDPR, audit trails

Risk tiers — prohibited, high-risk, transparency duties (Art. 50, since 2 Aug 2026), minimal — tiers can combine, not four disjoint boxes

Where agents land — by concrete use case and exceptions: hiring or credit decisions typically high-risk; a literature assistant is not

High-risk duties — risk management, logging, human oversight, robustness — AI Omnibus (in force 27 Jul 2026) defers to 2 Dec 2027 (Annex III) / 2 Aug 2028 (Annex I)

GPAI layer — model providers owe documentation and transparency since 2 Aug 2025; models placed earlier have until 2 Aug 2027

GDPR touchpoints — traces hold personal data: lawful basis, minimisation, erasure rights versus audit retention

Audit trail — auditability needs integrity, access control and the right content — S11 tracing alone is not a legal record

The footprint multiplier

100x the tokens of one call is not 100x the energy.

An agent pipeline can easily burn 100x the tokens of a single call: tens of loop steps, context re-sent each step, reasoning tokens, retries, parallel sub-agents; Anthropic reports multi-agent research runs near 15x chat token use. Tokens are an activity proxy: prefill, decoding, caching, hardware, utilisation and location decide the energy.

Where the watt-hours go

A measured anchor — Google: 0.24 Wh for the median Gemini Apps text prompt, May 2025 — per prompt, not per token, one product

Context regrowth — each loop step re-sends history — tokens processed grow quadratically with steps (S02)

Reasoning inflation — thinking models emit many more tokens for the same visible answer; measure your own ratio

Multi-agent multiplication — parallel sub-agents multiply everything again (S12)

What energy depends on — prefill vs decoding, cache hits, model and hardware, utilisation, grid location, tool work

Efficiency levers — cache prompts, route small-models-first, batch offline work, cap steps

PROXY, NOT METER

Tokens are an activity proxy, not an energy meter. Energy = attributed power integrated over time; emissions = energy x grid intensity, lifecycle shares separately. Measure where you can; state boundaries and uncertainty.

A responsible-deployment checklist

Six commitments before an agent touches production

Evals before autonomy

No agent widens its permissions without an eval set it passes: outcome checks on the real artifact, trajectory invariants, a held-out set, gates on every change.

Logs that answer audits

Every action traceable to its inputs and decision; tamper-evident, access-controlled, retention balanced against GDPR minimisation. What you cannot reconstruct, you cannot defend.

A named human owner

Each deployed agent has an accountable operator, an escalation path, and approval gates on irreversible actions (S10).

Least standing privilege

Scoped credentials, default-deny egress, ephemeral environments — the S10 and S13 machinery is a precondition, not an upgrade.

Budget every axis

Tokens, money, steps, energy. Cache, route small-first, batch offline work; alert on spend anomalies before the invoice does.

Plan for change

Models deprecate, prompts drift, regulation phases in. Re-run evals on every upgrade; keep a kill switch and a retirement plan.

The course in one picture

Nine units, one thread: a model in a loop with tools



- Each layer answers a failure mode of the previous one — capability created risk, risk demanded discipline.
- Lab arc: Lab 01 reflex agent and PEAS/BDI; Lab 02 minimal loop; Labs 03-09 tools, prompts, reasoning, orchestration, frameworks, memory, RAG; Labs 10-13 sandbox, traces, multi-agent, security.
- Lab 14 closes the arc: an eval harness that checks, judges, meters and gates what the earlier labs built — all on a local model.

The exam — and what to take with you

Format — written exam; details and scope in the exam-preparation notes

What we test — mechanisms and trade-offs: how it works, when it breaks, what it costs

Not memorisation — no API trivia, no framework syntax, no leaderboard numbers

How to practise — the ~50 questions per session are the rehearsal space — favour the design ones

Labs as evidence — fourteen notebooks on a local model: loop, tools, memory, RAG, sandbox, traces, eval harness

ONE SENTENCE

An agent is a model in a loop with tools. Everything else was making that loop capable, safe, observable, and accountable.

TAKEAWAYS

What you should remember

- **Evaluate in layers —** tool tests, step checks, trajectories, outcomes — statistics over runs
- **Judges are biased —** position, verbosity, self-preference — swap, keep inconsistent verdicts visible, calibrate
- **Benchmarks, honestly —** versions, budgets, contamination, benchmarketing; only your own eval set counts
- **Accountability is human —** provider, deployer, user under applicable law; auditable logs reconstruct the answer
- **Tokens are a proxy —** cache, route, batch, small-first; measure energy where you can and state the uncertainty
- **The course in one line —** a model in a loop with tools — made capable, then made trustworthy

NEXT SESSION Written exam — see the exam-preparation notes. Thank you, and good luck.

Reading & references

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